Tessa Adams
CariSurg Healthcare AI Program
Week 7 Decision Journal
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Decision Journal: Week 7 Model Choice

Context

The ED Board asked whether a more complex model provides enough benefit to justify its added cost, maintenance, and reduced interpretability.

Alternatives Considered

- Logistic Regression
- Tuned Random Forest
- Gradient Boosting
- Small MLP neural network
- Stacked Ensemble combining Logistic Regression and the Tuned Random Forest

Decision

I selected Logistic Regression as the primary model for Phase 3 and retained the tuned Random Forest as a comparison model.

Reasoning

Logistic Regression achieved the highest macro-F1 at 0.492, compared with 0.471 for the tuned Random Forest. The stacked ensemble improved ESI 1 recall to 0.69, but its ESI 1 precision was only 0.02 and its macro-F1 was lower at 0.440, so it was not selected. The more complex models did not improve overall performance enough to justify their additional training, explanation, and maintenance requirements.

Logistic Regression is also easier to interpret, deploy, and monitor in a clinical setting.

Remaining Questions

Further analysis is needed to determine:

- How safely the models handle ESI 1 and ESI 2 patients, particularly undertriage.
- Whether performance remains stable across different data splits, future patients, and patient groups.